



MCP Apps

Connect tool-result UI resources to an inline interactive preview.

Problem and boundary

Public contract

Working integration

@epam/ai-dial-mcp-apps



When this library is useful

Connect tool-result UI resources to an inline interactive preview.

01

Find a tool UI

Match a message to an MCP tool resource.

02

Reuse the result

Seed the preview from the original tool call.

03

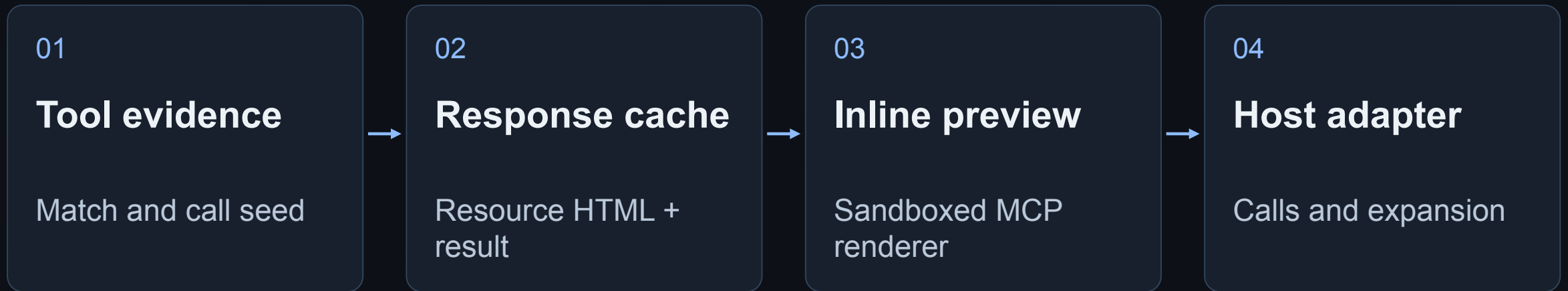
Expand the UI

Open the matched application in the attachment canvas.



Follow the ownership boundary

Inputs and effects stay explicit.



Library: Preview state, matching/seed helpers and a per-conversation response cache.



The public contract to start with

`McpAppInlinePreview` · selected surface; full API index is in the bundle.

Input or operation	Contract
match	<code>McpAppToolRef</code>
cache	<code>McpAppResponseCache</code>
hostAdapter	<code>McpAppHostAdapter</code>
toolCall (optional)	<code>McpAppToolCallSeed</code>



A minimal workspace integration

TypeScript / React example · imports and host setup abridged.

```
const cache = useMcpAppResponseCache('demo-conversation');  
<McpAppInlinePreview  
  match={match} cache={cache} cacheKey="demo-message"  
  hostAdapter={adapter} onExpand={onExpand}  
  expandAriaLabel="Expand demo app"  
  reloadAriaLabel="Reload"  
  loadErrorLabel="Could not load demo app"  
  openedInCanvasLabel="Open in preview"  
>
```

Full example: [examples/mcp-apps.tsx](#) · uses [@epam/ai-dial-mcp-apps](#)



Wire a realistic host workflow

Keep mcp apps focused on its public responsibility.

01

Match tool evidence

Provide the resource match and a stable cache key.

02

Inject the host adapter

The host fetches resources and handles tool calls.

03

Reuse cached state

Keep response caching scoped to the conversation lifetime.



Available customization

01

Explicit host adapter

Inject
fetchResourceHtml,
callTool, hostInfo and
sandboxUrl.

02

Cache identity

Entries are checked
against seed identity
and expiry.

03

Lifecycle state

Loading, Ready, Error
and Unavailable are
distinct enum values.



Find the integration and its checks

Existing tests are evidence of intended behavior; suite results are not claimed.

Evidence	Where to look
Real consumer	chat-hooks/src/mcp-apps/useMcpAppTools/useMcpAppTools.ts
Test evidence	stamps the preview card and its header



Constraints that affect integration

01

Origin isolation required

Configure the separate MCP sandbox deployment.

02

Avoid duplicate side effects

Indirectly discovered tools are not safe to recall blindly.

03

Not a general API client

Backend operations enter through the host adapter.